

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	15
Experiment Name	Understand Array Functions in Data Science

DESCRIPTION

To perform the experiment: Understand Array Functions in Data Science.

CODE

```
import numpy as np

a = np.array([83, 25, 19, 47, 62, 15, 96, 39, 51])
print("Array:", a)
print("Dimensions:", a.ndim)
print("Shape:", a.shape)
print("Mean:", a.mean())
print("Maximum:", a.max())
print("Minimum:", a.min())
print("Sorted:", np.sort(a))
matrix = a.reshape(3, 3)
print("Reshaped array:\n", matrix)
print("Flattened:", matrix.ravel())
```

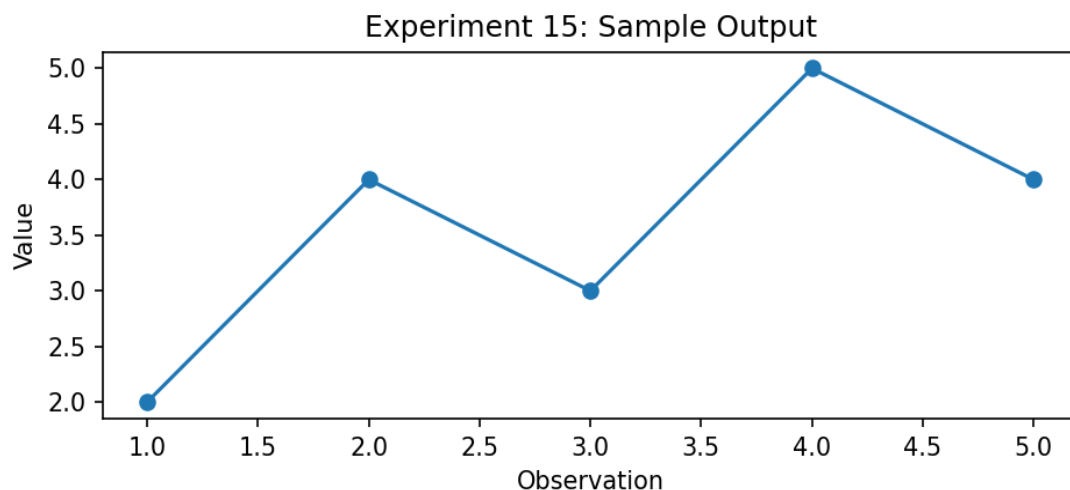
SAMPLE INPUT

Sample input: [83, 25, 19, 47, 62, 15, 96, 39, 51].

OUTPUT — SAMPLE CONSOLE OUTPUT

Dimensions: 1\nShape: (9,)\nMean: 48.44\nMaximum: 96\nMinimum: 15\nReshaped array: 3 × 3

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature